1. (a) $-\frac{2}{3}(\cos(x))^{3/2}+C$

   (b) $2\sin^{-1}\left(\frac{x+1}{2}\right)+\frac{x+1}{2}\sqrt{3-2x-x^2}+C$

2. (a) 4

   (b) $\frac{1}{2}\ln(2)-\frac{\pi}{12}$

3. $\frac{35+9\pi}{18+3\pi}$

4. (a) Converges to 2

   (b) Diverges

5. $\frac{12250}{3}\pi R^4$ Joules

6. (a) $\int_1^4 \sqrt{1+25x^3}\,dx$

   (b) $\frac{3}{8}\cdot\left(\sqrt{1+25}+2\sqrt{1+25\cdot\left(\frac{7}{4}\right)^3}+2\sqrt{1+25\cdot\left(\frac{10}{4}\right)^3}+2\sqrt{1+25\cdot\left(\frac{13}{4}\right)^3}+\sqrt{1+25\cdot(4)^3}\right)$

7. (a) $\frac{\pi}{4}$

   (b) $\frac{\pi^3}{8}+\frac{\pi}{2}$

8. (a) $\frac{1}{6}\cdot(7+2\pi)$

   (b) 4

9. $y=-\sqrt{9+\sin(x^2)}$

10. $t=\frac{4\ln 29}{\ln 29-\ln 5}\approx 7.66$